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Editor-in-Chief
International Journal of Control, Automation, and Systems

Dear Editor-in-Chief,

Please consider the manuscript entitled "Complementary-Filtered Residual World Models for Receding-Horizon Control of Nonlinear Systems" for publication in the *International Journal of Control, Automation, and Systems*.

The manuscript studies a state-estimation and predictive-control problem in which forecast-origin observation error and transition error accumulate over a receding horizon. It introduces a complementary-filtered residual world model that combines a mask-gated context update, nominal dynamics, and a bounded neural residual. The evaluation comprises five nonlinear controlled systems, a frozen 1,920-episode receding-horizon control study, and held-out measured NanoDrone flight records.

The results distinguish the supported contribution from the components that do not receive additional empirical support. The complete model improves on the preselected physics-guided neural comparator in the aggregate controlled benchmark and in realized closed-loop state loss. At the same time, clean-regime noninferiority is not established, filtering followed by nominal dynamics slightly outperforms the complete model, and the measured-flight study exposes higher full-state error and undercoverage despite lower long-horizon position error. These findings are relevant to the journal's scope in control engineering, automation, and artificial intelligence for engineered systems because they identify the practical value of forecast-origin filtering while delimiting the contribution of the learned residual and uncertainty output.

The manuscript contains the author information, data and code availability statement, funding statement, competing-interest declaration, author contribution statement, and applicable ethics and consent statements. Complete editable LaTeX sources and figure files accompany the submission.

Thank you for considering this manuscript.

Sincerely,

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